

Supplementary Materials for “Non-Abelian statistics of Majorana zero modes in the presence of an Andreev bound state”

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I. PROTOCOL FOR THE BRAIDING OPERATION

An illustration of the braiding protocol is shown in Fig. S1(a). The MZMs are moved by modulating the hopping strengths between the four arms and the trivial superconductor through the gate voltages shown in Fig. S1(b). Such electron hopping strengths t_{ic} can be modulated by gates G_i as

$$t_{ic} = g_i t_0, \quad (S1)$$

where g_i is the gate voltage in G_i and in the time-dependent form as Fig. S1(b). As we assume that the chemical potential μ_c as well as the hopping amplitude t_0 are uniform inside the tetron structure, the i th wire is perfectly connected to the trivial superconductor when $g_i = 1$. Initially before the braiding, we set $g_{1,2} = 0$ and $g_{3,4} = 1$ so that three pairs of MZMs (γ_{2j-1} and γ_{2j} , with $j = 1, 2, 3$) are localized at the present domain walls. The effective low-energy Hamiltonian to describe the configuration of this starting point is

$$H_{eff} = iE_1\gamma_1\gamma_2 + iE_2\gamma_3\gamma_4 + iE_3\gamma_5\gamma'_5 + iE_4\gamma_6\gamma'_6 \\ + it_c(g_1g_2\gamma_2\gamma_3 + g_1g_3\gamma'_5\gamma_2 + g_1g_4\gamma_2\gamma'_6 + g_2g_3\gamma_3\gamma'_5 + g_2g_4\gamma_3\gamma'_6 + g_3g_4\gamma'_5\gamma'_6) + h.c.. \quad (S2)$$

Note that γ'_5 and γ'_6 are in fact not in the zero-energy subspace, because they are couple to each other that they are actually the high-energy modes that are above the topological gap. However, γ'_5 and γ'_6 are introduced here as two “auxiliary MZMs” located near the trivial superconductor in order to facilitate our understanding of the movement of the actual zero modes. For simplicity,

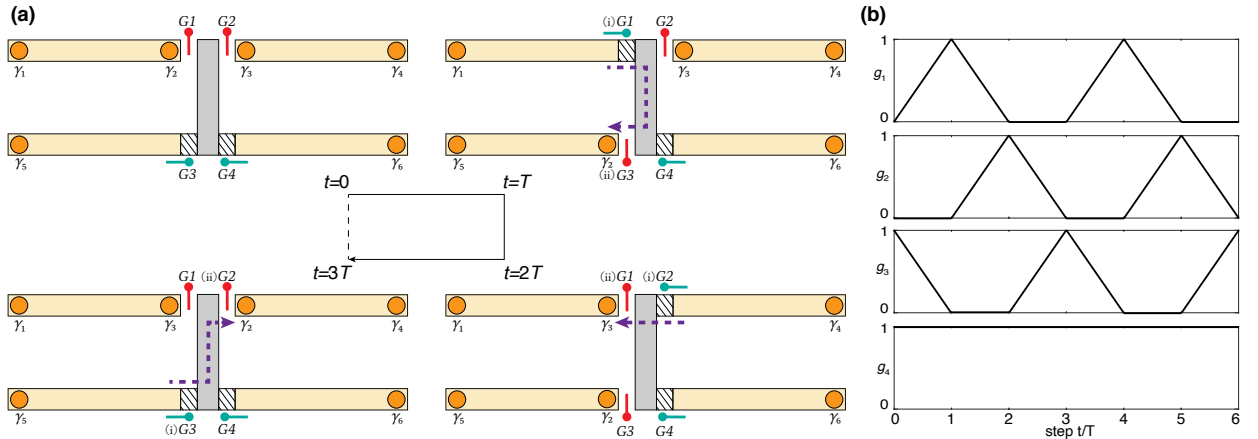


FIG. S1: Schematic plot of the braiding operation. (a) The six-step braiding protocol. After the first three steps ($t = 3T$), the spatial positions of two MZMs γ_2 and γ_3 are swapped. The last three steps ($t \in [0, 6T]$) repeat the operation in the first three steps. (b) Gate voltage g_i 's as functions of the braiding step. The hopping strengths between the nanowires and the trivial superconductor are modulated by the corresponding gate voltages, therefore they have the same time-dependence as g_i . Here the g_i 's are linearly turned on or turned off, while other forms of the g_i functions such as sinusoids should also work well.

we assume that the tunnel couplings between any two MZMs near the junction are approximately identical, represented by t_c , whereas the coupling between γ'_5 and γ'_6 is larger. The connectivity of the gates is specified by a parameter set (g_1, g_2, g_3, g_4) . The finite-size effects are exponentially small compared to the coupling t_c at the junction, i.e. $E_i \rightarrow 0$, at the limit that $N_x \gg N_c$, so we can safely neglect the first four terms in the above effective Hamiltonian,

$$H_{eff} = it_c(g_1 g_2 \gamma_2 \gamma_3 + g_1 g_3 \gamma'_5 \gamma_2 + g_1 g_4 \gamma_2 \gamma'_6 + g_2 g_3 \gamma_3 \gamma'_5 + g_2 g_4 \gamma_3 \gamma'_6 + g_3 g_4 \gamma'_5 \gamma'_6) + h.c.. \quad (S3)$$

Moving a MZM from the i th wire to the j th wire is accomplished by adiabatically switching on g_i ($t \in [0, T/2]$) and then switching off g_j ($t \in [T/2, T]$). After such step of gate operation (time-cost is T), the MZM will tunnel through the trivial superconductor and move to another nanowire [indicated by the dashed arrow in Fig. S1(a)]. The braiding protocol takes three steps (the time-cost for each step is T) to swap γ_2 and γ_3 spatially. In the following we show the detailed procedure of each step by tracking the instantaneous zero-energy modes γ_θ and γ_ϕ of the effective low-energy Hamiltonian. The reason for the θ, ϕ notation will be clear by the end of this section. At the adiabatic limit, which is $1/T$ is very small compared to the topological gap, these instantaneous zero modes are the same as the time-evolved MZMs from their initial states. In step 1, G_2 is kept closed ($g_2 = 0$), so H_{eff} can be further simplified to $H_{eff} = it_c(g_1 g_3 \gamma'_5 \gamma_2 + g_1 g_4 \gamma_2 \gamma'_6 + g_3 g_4 \gamma'_5 \gamma'_6) + h.c..$ By diagonalizing H_{eff} , the two instantaneous zero-energy modes are

$$\gamma_\theta = \frac{1}{C}(\frac{1}{g_1} \gamma_2 + \frac{1}{g_3} \gamma'_5 + \frac{1}{g_4} \gamma'_6) \quad (S4)$$

and

$$\gamma_\phi = \gamma_3 \quad (S5)$$

The gate G_1 is turned off and then G_3 is turned on, so the gate set (g_1, g_2, g_3, g_4) is switched from $(0, 0, 1, 1)$ to $(1, 0, 0, 1)$. We can see that γ_θ evolves from γ_2 to γ'_5 , which shows that γ_2 is teleported through the trivial superconductor to W3. In step 2, G_3 is kept closed ($g_3 = 0$). The two instantaneous zero-energy modes are

$$\gamma_\theta = \gamma'_5 \quad (S6)$$

and

$$\gamma_\phi = \frac{1}{C}(-\frac{1}{g_1} \gamma_2 + \frac{1}{g_2} \gamma_3 + \frac{1}{g_4} \gamma'_6) \quad (S7)$$

It is noteworthy that the minus sign comes naturally from diagonalizing H_{eff} . G_2 is turned off and then G_1 is turned on, so the gate set (g_1, g_2, g_3, g_4) is switched from $(1, 0, 0, 1)$ to $(0, 1, 0, 1)$, then γ_θ evolves from γ_3 to $-\gamma_2$. This shows that γ_3 is teleported through the trivial superconductor to W1 and obtains a π -phase. Finally in step 3, G_1 is kept closed ($g_1 = 0$). The two instantaneous zero-energy modes are

$$\gamma_\theta = \frac{1}{C}(\frac{1}{g_2} \gamma_3 + \frac{1}{g_3} \gamma'_5 + \frac{1}{g_4} \gamma'_6) \quad (S8)$$

and

$$\gamma_\phi = -\gamma_2 \quad (S9)$$

G_3 is turned off and then G_2 is turned on. The gate set (g_1, g_2, g_3, g_4) is switched from $(0, 1, 0, 1)$ to $(0, 0, 1, 1)$, then we can see that γ_θ evolves from γ'_5 to γ_3 , which shows that γ_2 is teleported through the trivial superconductor to W2. Consequently, the spatial positions of γ_2 and γ_3 are swapped.

In the braiding protocol, G_4 is kept off ($g_4 = 1$) throughout the procedure while $G_1 \sim G_3$ are turned off one by one each step in a cyclic way ($\{g_1, g_2, g_3\} = \{0, 0, 1\} \rightarrow \{1, 0, 0\} \rightarrow \{0, 1, 0\} \rightarrow \{0, 0, 1\}$), thus the instantaneous zero modes are never equal to γ'_6 . When diagonalizing H_{eff} , we know that the $\frac{1}{g_4} \gamma'_6$ comes from the $it_c g_1 g_3 \gamma'_5 \gamma_2$ term in H_{eff} in step 1 (or $it_c g_1 g_2 \gamma_2 \gamma_3$ in step 2, $it_c g_2 g_3 \gamma'_5 \gamma_3$ in step 3). These terms don't contribute to the movement of the instantaneous zero modes, and therefore, can be dropped from the effective Hamiltonian. Through this argument, the Hamiltonian could further be simplified to an elegant form

$$H = i\gamma'_6(\vec{\delta} \cdot \vec{\gamma}) \quad (S10)$$

where $\vec{\gamma} = (\gamma_2, \gamma_3, \gamma'_5)$ and $\vec{\delta} = (t_1, t_2, t_3) = t_c(g_1 g_4, g_2 g_4, g_3 g_4)$. Hence the instantaneous eigenstates are MZM $\gamma_2 = \vec{\gamma} \cdot \hat{e}_\theta$, denoted as the vector along the polar direction (θ -direction) perpendicular to $\vec{\delta}$, and MZM $\gamma_3 = \vec{\gamma} \cdot \hat{e}_\phi$, denoted as the vector along the polar direction (θ -direction) perpendicular to $\vec{\delta}$.

II. INFLUENCE OF THE DISTANCE BETWEEN TWO PARALLEL SEMICONDUCTOR-SUPERCONDUCTING NANOWIRE

The modified “tetron” qubit is connected through a trivial superconductor. In this case, the length of the trivial superconductor is hard to be short. Here we study the effect induced by the length of the trivial superconductor. Fig. S2 shows the evolution of $\psi_1^+(t)$ with different lengths of the trivial superconductor N_c as $N_c = 6$ in Fig. S2(a), and $N_c = 9$ in Fig. S2(b). We can see that the non-Abelian braiding properties are well preserved in both these two cases. The corresponding evolution of the eigenenergy spectrum during the braiding process is shown in Fig. S2(c) and Fig. S2(d), respectively. We can see that the bulk gap decreases with the increase of N_c . However, the gap remains open even when N_c is in the order of the superconducting coherence length. This means that the non-Abelian braiding properties can be well preserved when the length of the trivial superconductor N_c is smaller than the superconducting coherence length.

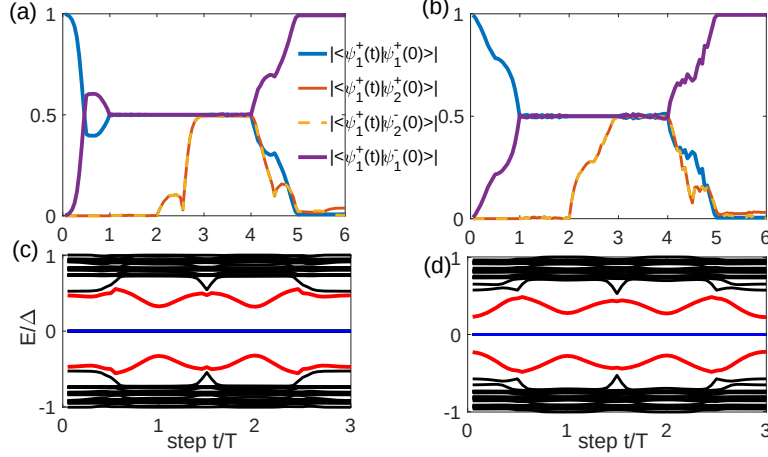


FIG. S2: (a), (b) Non-Abelian braiding of the modified “tetron” qubit with length of the trivial superconductor $N_c = 6$ in (a), and $N_c = 9$ in (b). (c), (d) The corresponding eigenenergy spectra during the braiding process with length $N_c = 6$ in (c), and $N_c = 9$ in (d).

III. SUPERCONDUCTING PHASE KNOB FOR THE TETRON STRUCTURE

Because the tetron qubit consists of two parallel-aligned nanowires, a domain wall structure will be formed at the position that $W2$ and $W4$ intersect the trivial superconductor when $t = 2T$. Figure S3(a) shows that the energy of the bound states decrease to zero at $t = 2T$, indicating that an additional pair of zero modes will emerge from the bulk at the domain wall. These two extra modes will cause information leakage to the bulk states and therefore ruin the braiding [see Fig. S1(b)]. Fortunately, a superconducting phase knob is provided to preserve the robustness of the non-Abelian braiding. By tuning the superconducting phase difference ϕ_0 to a non-zero value, the domain wall could be smoothed out. Hence the energy gap remains open throughout the braiding process [Fig. 1(e) in the main text] and the non-Abelian braiding properties remain [Fig. 1(f) in the main text].

Superconducting phase can smooth out the domain wall structure and enlarge the bulk gap. Here we study the variance of the bulk gap as a function of the superconducting phase. Since the bulk gap is minimized when the nanowires are fully connected, we consider the variance of the bulk gap versus the superconducting phase in the different connection conditions of the voltage gates. Fig. S4 shows the lowest four eigenenergies versus the superconducting phase difference ϕ_0 . Here, G1, G3, and G4 are connected when $\phi_0 \in (0, 2\pi)$; G1, and G4 are connected when $\phi_0 \in (2\pi, 4\pi)$; G1, G2, and G4 are connected when $\phi_0 \in (4\pi, 6\pi)$; G2 and G4 are connected when $\phi_0 \in (6\pi, 8\pi)$; G2, G3, and G4 are connected when $\phi_0 \in (8\pi, 10\pi)$; G3 and G4 are connected when $\phi_0 \in (10\pi, 12\pi)$. We can see that the bulk gap is optimized if $\phi_0 = 0$ in the time period $t \in (0, T)$ and $t \in (2T, 3T)$, while $\phi_0 = \pi$ in the time period $t \in (T, 2T)$.

IV. EVOLUTION OF THE LOCAL DENSITY OF STATES

The non-Abelian braiding properties of the MZMs is independent of the braiding dynamics. The wave functions evolve into a fixed form in a robust way at each step and are independent of the braiding time-cost T . However, the presence of the ABS make the braiding susceptible to the dynamic effect. We plot the local density of states (LDOS) distribution of both the ABS and the

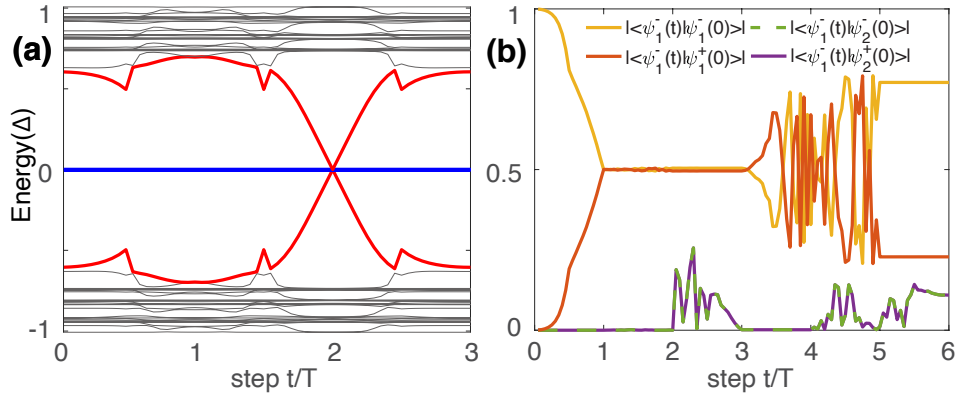


FIG. S3: (a) The eigenenergy spectrum as a function of time during the braiding process when $\phi_0 = 0$. The gap is closed at the end of the second braiding step ($t = 2T$). (b) Evolution of $\psi_1^-(t)$ during the braiding process when $\phi_0 = 0$. The braiding fails to exhibit non-Abelian properties in such case.

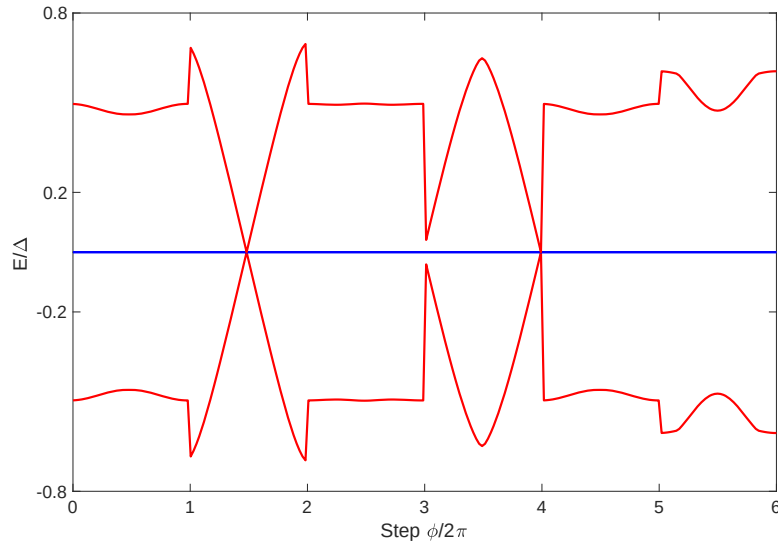


FIG. S4: The lowest four eigenvalues versus the superconducting phase in different connect setup. Here G1, G3, and G4 are connected when $\phi_0 \in (0, 2\pi)$; G1 and G4 are connected when $\phi_0 \in (2\pi, 4\pi)$; G1, G2, and G4 are connected when $\phi_0 \in (4\pi, 6\pi)$; G2 and G4 are connected when $\phi_0 \in (6\pi, 8\pi)$; G2, G3, and G4 are connected when $\phi_0 \in (8\pi, 10\pi)$; G3 and G4 are connected when $\phi_0 \in (10\pi, 12\pi)$.

MZMs at $t = T$, $t = 5T$, and $t = 6T$ as functions of T in Fig. S5. At $t = T$, the LDOS splits into two parts with equal weight. It means that two MZMs split from the ABS regardless of the value of T . At $t = 5T$, part of ABS gradually oscillates from $W1$ to $W3$ as T varies from $400/\Delta$ to $450/\Delta$. At the same time, part of the MZMs on $W3$ oscillates to $W1$. The ABS and the MZMs at $t = 6T$ have the same pattern and exhibit an oscillation with respect to T . The LDOS in Fig. 2 in the main text and Fig. S5 reveals that during the braiding, one of the MZMs that initially bounded in the ABS will split from the other MZM. After that, another MZM will fuse with the first MZM to form a new ABS. Such fusion and splitting behavior of the ABS provides further evidence supporting the twin-peak hypothesis that an ABS consists of two coupled MZMs. Moreover, the T -dependence is also a firm evidence for the dynamical hybridization of the MZMs in the ABS. Due to the non-zero coupling energy, the MZMs hybridize and thus lead to sinusoidal oscillation as shown in the LDOS distributions. Based on the understanding of the fusion and the hybridization, we can fully understand the braiding behaviors of the ABS and predict the evolution of the wave functions during the braiding.

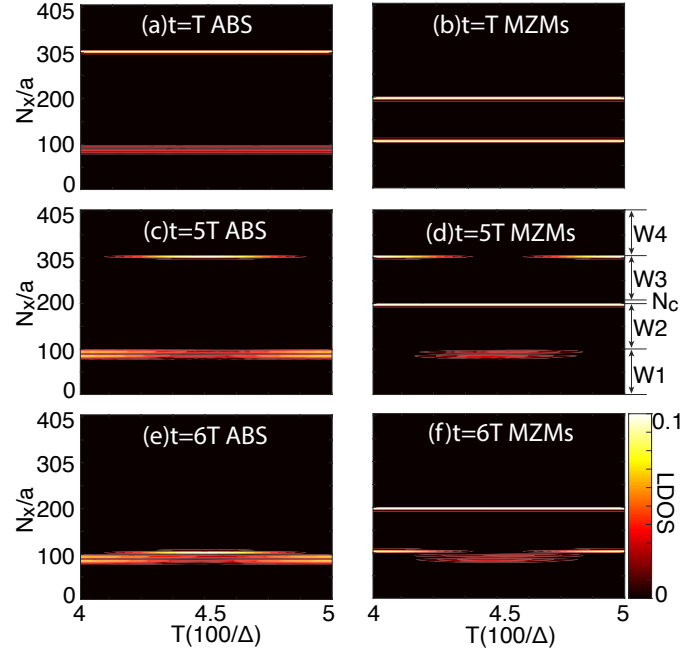


FIG. S5: The LDOS at $t = T$, $t = 5T$, and $t = 6T$ as functions of the time period T shows that the spatial distribution of the ABS and the MZMs oscillates with the braiding time-cost T . At $t = T$, the LDOS splits into two parts with equal weight, which indicates that the two MZMs split from the ABS. While at $t = 5T$ and $t = 6T$, The LDOS displays sinusoidal oscillation behavior. Such sinusoidal dependence is a signature of the hybridization between one “free” MZM and another MZM bounded in the ABS.

V. EVOLUTION OF THE WAVE FUNCTIONS DURING THE BRAIDING

The effective low-energy Hamiltonian for the modified “tetron” throughout the braiding takes the form of

$$H(t) = i \sum \epsilon_{m,n}(t) \gamma_m(t) \gamma_n(t). \quad (\text{S11})$$

Figure 3(a) in the main text shows the hybridization energies $\epsilon_{m,n}(t)$ of three different pairs of MZMs in the “tetron” as functions of t during the braiding. In Fig. 3(a) in the main text, the blue and red curves are the hybridization energies of the MZMs which fuse and form the ABS, while the zero-energy black line characterizes the rest well-separated MZMs. The evolution operator is calculated by plugging in $H(t)$ to $\exp\left(i \int_0^t d\tau H(\tau)\right)$. This unitary evolution includes two parts: the dynamical part and the topological part. The dynamical part has an evolution operator in the form of $\exp\left(\frac{\theta(t)}{2} \gamma_m \gamma_n\right)$ in which $\theta(t)/2 = \int_0^t d\tau \epsilon_{m,n}(\tau)$, while the corresponding $\theta/2$ in the topological part is fixed at $\pi/4$. During the braiding process in which an ABS is also involved, both the dynamical evolution and the topological evolution alternatively perform. Hence, the whole braiding operator is in the form of the product of these dynamical and the topological evolution operators.

To be more specific, in this section, we are going to derive both the unitary transformations on the Majorana operators and the wave functions in each step of the gate operation. As shown in Fig. S1, the last three operation steps repeat the operation in the first three steps. Therefore, we only need to derive the evolution operator for the first three braiding steps. In the first step during $t \in [0, T]$, the hybridization energy $\epsilon_{(1)} \neq 0$ during the splitting process of γ_1 and γ_2 . Hence the evolution operator reads

$$U_{(1)} = \exp\left(\frac{\theta_{(1)}}{2} \gamma_1 \gamma_2\right). \quad (\text{S12})$$

where $\theta_{(1)}/2 = \int_0^T \epsilon_{(1)}(t) dt$. It can be easily shown that such evolution operator will lead to the hybridization between γ_1 and γ_2 :

$$\begin{aligned} \gamma_1^{(1)} &= U_{(1)} \gamma_1 U_{(1)}^\dagger = \cos(\theta_{(1)}) \gamma_1 + \sin(\theta_{(1)}) \gamma_2, \\ \gamma_2^{(1)} &= U_{(1)} \gamma_2 U_{(1)}^\dagger = -\sin(\theta_{(1)}) \gamma_1 + \cos(\theta_{(1)}) \gamma_2. \end{aligned} \quad (\text{S13})$$

Through the relationship $\psi_1^- = 1/\sqrt{2}(\gamma_1 + i\gamma_2)$, $\psi_1^+ = 1/\sqrt{2}(\gamma_1 - i\gamma_2)$, $\psi_2^- = 1/\sqrt{2}(\gamma_3 + i\gamma_4)$, $\psi_2^+ = 1/\sqrt{2}(\gamma_3 - i\gamma_4)$, such evolution operator $U_{(1)}$ is also equivalent to a unitary transformation matrix on the basis of the initial wave functions $\{|\psi_1^-\rangle, |\psi_1^+\rangle, |\psi_2^-\rangle, |\psi_2^+\rangle\}^T$:

$$U_{(1)} = \begin{bmatrix} e^{-i\theta_{(1)}} & 0 & 0 & 0 \\ 0 & e^{i\theta_{(1)}} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}. \quad (\text{S14})$$

At $t = T$, $\gamma_2^{(1)}$ fully splits from $\gamma_1^{(1)}$. After that, $\gamma_3^{(1)}$ starts to fuse with $\gamma_1^{(1)}$. The branch cut of the MZMs lead to a topological evolution operator

$$B = \exp\left(\frac{\pi}{4}\gamma_2^{(1)}\gamma_3^{(1)}\right), \quad (\text{S15})$$

which braids $\gamma_2^{(1)}$ and $\gamma_3^{(1)}$ as

$$\begin{aligned} \gamma_2^{(1)} &\rightarrow B\gamma_2^{(1)}B^\dagger = \gamma_3^{(1)}, \\ \gamma_3^{(1)} &\rightarrow B\gamma_3^{(1)}B^\dagger = -\gamma_2^{(1)}. \end{aligned} \quad (\text{S16})$$

Now the initial wave functions are transformed by

$$B \cdot U_{(1)} = \begin{bmatrix} \frac{1}{2}e^{-i\theta_{(1)}} & \frac{1}{2}e^{i\theta_{(1)}} & \frac{i}{2} & \frac{i}{2} \\ \frac{1}{2}e^{-i\theta_{(1)}} & \frac{1}{2}e^{i\theta_{(1)}} & -\frac{i}{2} & -\frac{i}{2} \\ \frac{i}{2}e^{-i\theta_{(1)}} & -\frac{i}{2}e^{i\theta_{(1)}} & \frac{1}{2} & -\frac{1}{2} \\ \frac{i}{2}e^{-i\theta_{(1)}} & -\frac{i}{2}e^{i\theta_{(1)}} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}. \quad (\text{S17})$$

Henceforth $\gamma_1^{(1)}$ and $\gamma_3^{(1)}$ are fused. In the second braiding step, the hybridization energy between $\gamma_1^{(1)}$ and $\gamma_3^{(1)}$ is non-zero, thus the corresponding hybridization operator

$$U_{(2)} = \exp\left(\frac{\theta_{(2)}}{2}\gamma_1^{(1)}\gamma_3^{(1)}\right), \quad (\text{S18})$$

transforms $\gamma_1^{(1)}$ and $\gamma_3^{(1)}$ as

$$\begin{aligned} \gamma_1^{(2)} &= U_{(2)}\gamma_1^{(1)}U_{(2)}^\dagger = \cos(\theta_{(2)})\gamma_1^{(1)} + \sin(\theta_{(2)})\gamma_3^{(1)}, \\ \gamma_3^{(2)} &= U_{(2)}\gamma_3^{(1)}U_{(2)}^\dagger = -\sin(\theta_{(2)})\gamma_1^{(1)} + \cos(\theta_{(2)})\gamma_3^{(1)}. \end{aligned} \quad (\text{S19})$$

$\gamma_1^{(1)}$ and $\gamma_3^{(1)}$ continues to hybridize in the third step. Therefore, after the second and the third braiding step, the total evolution operator can be expressed in the form of a transformation matrix as

$$U_{(3)} \cdot U_{(2)} \cdot B \cdot U_{(1)} = \begin{bmatrix} \frac{1}{2}e^{i(-\theta_{(1)}-\theta_{(2)}-\theta_{(3)})} & \frac{1}{2}e^{i(\theta_{(1)}-\theta_{(2)}-\theta_{(3)})} & \frac{i}{2}e^{i(-\theta_{(2)}-\theta_{(3)})} & \frac{i}{2}e^{i(-\theta_{(2)}-\theta_{(3)})} \\ \frac{1}{2}e^{i(-\theta_{(1)}+\theta_{(2)}+\theta_{(3)})} & \frac{1}{2}e^{i(\theta_{(1)}+\theta_{(2)}+\theta_{(3)})} & -\frac{i}{2}e^{i(\theta_{(2)}+\theta_{(3)})} & -\frac{i}{2}e^{i(\theta_{(2)}+\theta_{(3)})} \\ \frac{i}{2}e^{-i\theta_{(1)}} & -\frac{i}{2}e^{i\theta_{(1)}} & \frac{1}{2} & -\frac{1}{2} \\ \frac{i}{2}e^{-i\theta_{(1)}} & -\frac{i}{2}e^{i\theta_{(1)}} & -\frac{1}{2} & \frac{1}{2} \end{bmatrix}. \quad (\text{S20})$$

The projections of the wave functions at $t = 3T$ on the initial ones are all $1/2$. Operator $U_{(3)} \cdot U_{(2)} \cdot B \cdot U_{(1)}$ represents the first half of braiding procedure. The last three steps repeat the operations during the first three steps in $t \in [0, 3T]$, for both the hybridization energy [Fig. 3(a) in the main text] and the gate voltage profile [Fig. S1(b)] being in the exact same form in $t \in [0, 3T]$ and $t \in [3T, 6T]$. Therefore, the whole braiding process can be characterized by the evolution operator of

$$U_F = (U_{(3)} \cdot U_{(2)} \cdot B \cdot U_{(1)})^2 \quad (\text{S21})$$

In the basis of the initial wave functions, the matrix representation of U_F is

$$U_F = \begin{bmatrix} \frac{1}{2}e^{i(-\theta_1-\theta_3)}(\cos\theta_2-1) & \frac{1}{2}e^{i(\theta_1-\theta_3)}(\cos\theta_2+1) & \frac{1}{2}e^{-i\theta_3}\sin\theta_2 & \frac{1}{2}e^{-i\theta_3}\sin\theta_2 \\ \frac{1}{2}e^{i(-\theta_1+\theta_3)}(\cos\theta_2+1) & \frac{1}{2}e^{i(\theta_1+\theta_3)}(\cos\theta_2-1) & \frac{1}{2}e^{i\theta_3}\sin\theta_2 & \frac{1}{2}e^{i\theta_3}\sin\theta_2 \\ \frac{1}{2}e^{-i\theta_1}\sin\theta_2 & \frac{1}{2}e^{i\theta_1}\sin\theta_2 & \frac{1}{2}(1-\cos\theta_2) & \frac{1}{2}(1+\cos\theta_2) \\ \frac{1}{2}e^{-i\theta_1}\sin\theta_2 & \frac{1}{2}e^{i\theta_1}\sin\theta_2 & \frac{1}{2}(1+\cos\theta_2) & \frac{1}{2}(1-\cos\theta_2) \end{bmatrix}, \quad (\text{S22})$$

where $\theta_1 = \theta_{(1)}, \theta_2 = \theta_{(2)} + \theta_{(3)} + \theta_{(4)}, \theta_3 = \theta_{(5)} + \theta_{(6)}$. Such results show that, though the braiding protocol is composed of six steps of gate voltage operations, it can also be divided into five evolutionary sections according to the time-dependence of hybridization energies. These five sections are divided as follows: Section 1. $\exp(\frac{\theta_1}{2}\gamma_1\gamma_2)$, i.e., hybridization of γ_1 and γ_2 ; Section 2. $\exp(\frac{\pi}{4}\tilde{\gamma}_2\gamma_3)$, i.e., braiding of $\tilde{\gamma}_2$ and γ_3 ; Section 3. $\exp(\frac{\theta_2}{2}\tilde{\gamma}_1\gamma_3)$, i.e., hybridization of $\tilde{\gamma}_1$ and γ_3 ; Section 4. $\exp(-\frac{\pi}{4}\tilde{\gamma}_3\tilde{\gamma}_2)$, i.e., braiding of $\tilde{\gamma}_3$ and $-\tilde{\gamma}_2$; Section 5. $\exp(-\frac{\theta_3}{2}\tilde{\gamma}_1\tilde{\gamma}_2)$, i.e., hybridization of $\tilde{\gamma}_1$ and $-\tilde{\gamma}_2$. Here, $\tilde{\gamma}_1 = \exp(\frac{\theta_1}{2}\gamma_1\gamma_2)\gamma_1\exp(-\frac{\theta_1}{2}\gamma_1\gamma_2)$, $\tilde{\gamma}_1 = \exp(\frac{\theta_2}{2}\tilde{\gamma}_1\gamma_3)\tilde{\gamma}_1\exp(-\frac{\theta_2}{2}\tilde{\gamma}_1\gamma_3)$, $\tilde{\gamma}_2 = \exp(\frac{\theta_1}{2}\gamma_1\gamma_2)\gamma_2\exp(-\frac{\theta_1}{2}\gamma_1\gamma_2)$, and $\tilde{\gamma}_3 = \exp(\frac{\theta_2}{2}\tilde{\gamma}_1\gamma_3)\gamma_3\exp(-\frac{\theta_2}{2}\tilde{\gamma}_1\gamma_3)$. These notations indicate that the accumulative behaviors of MZMs cannot be neglected and every section of the evolution is based on the final states of the last section. Therefore, the evolution operator should be represented by a matrix that is expanded on these final states. If the expansion of the matrix for each evolution section is based on the initial states, i.e., the accumulative effect is neglected, then the operator for the complete evolution will be a direct product of the braiding and hybridization operator as $U'_F = \exp(\frac{\theta_3}{2}\gamma_1\gamma_2)\exp(\frac{\pi}{4}\gamma_2\gamma_3)\exp(\frac{\theta_2}{2}\gamma_1\gamma_3)\exp(\frac{\pi}{4}\gamma_2\gamma_3)\exp(\frac{\theta_1}{2}\gamma_1\gamma_2)$. Based on the same initial wave functions, the matrix representation then becomes

$$U'_F = \begin{bmatrix} 0 & e^{i\theta_1}e^{-i\theta_2}e^{-i\theta_3} & 0 & 0 \\ e^{-i\theta_1}e^{i\theta_2}e^{i\theta_3} & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & -1 & 0 \end{bmatrix}. \quad (\text{S23})$$

In this case, the physics effect of U_F is the same as the non-Abelian braiding of MZMs without ABS, where an additional unobservable phase factor oscillates with T . Such argument proves the accumulative effects in the ABS-engaged braiding.

Finally, by transforming to the many-body basis $\{|0\rangle, \Psi_1^\dagger|0\rangle, \Psi_2^\dagger|0\rangle, \Psi_1^\dagger\Psi_2^\dagger|0\rangle\}$, the complete evolution operator during the braiding is represented by

$$U_F = i \begin{bmatrix} e^{i\frac{\theta_1+\theta_3}{2}}S_{\frac{\theta_2}{2}} & 0 & 0 & e^{i\frac{\theta_1-\theta_3}{2}}C_{\frac{\theta_2}{2}} \\ 0 & -e^{-i\frac{\theta_1+\theta_3}{2}}S_{\frac{\theta_2}{2}} & e^{-i\frac{\theta_1-\theta_3}{2}}C_{\frac{\theta_2}{2}} & 0 \\ 0 & e^{i\frac{\theta_1-\theta_3}{2}}C_{\frac{\theta_2}{2}} & -e^{i\frac{\theta_1+\theta_3}{2}}S_{\frac{\theta_2}{2}} & 0 \\ e^{-i\frac{\theta_1-\theta_3}{2}}C_{\frac{\theta_2}{2}} & 0 & 0 & e^{-i\frac{\theta_1+\theta_3}{2}}S_{\frac{\theta_2}{2}} \end{bmatrix}, \quad (\text{S24})$$

where $S_{\frac{\theta_2}{2}} \equiv \sin \frac{\theta_2}{2}$ and $C_{\frac{\theta_2}{2}} \equiv \cos \frac{\theta_2}{2}$.

VI. NUMERICAL SIMULATION OF THE BRAIDING RESULTS FOR OTHER STATES

In the last section, we have shown the analytical results of the non-Abelian braiding of MZMs with an ABS engaged. The final braiding results can be described by Eq. (S22) and Eq. (S24). We can see that the final braiding operator show similar behavior for the different wave functions. In main text, we take $\psi_1^-(6T)$ as an example to display the oscillation behavior with respect to the braiding time-cost T . Here we show the numerical simulation of the braiding results as functions of T for other states as $\psi_1^+(6T)$ in Fig. S6 (a), $\psi_2^-(6T)$ in Fig. S6 (b), and $\psi_2^+(6T)$ in Fig. S6 (c), respectively. The projection of $|\langle\psi_1^+(6T)|\psi_1^-(0)\rangle|$, $|\langle\psi_2^-(6T)|\psi_2^-(0)\rangle|$, and $|\langle\psi_2^+(6T)|\psi_2^-(0)\rangle|$ in Fig. S6 can all be well described by the same formula as $(1 + \cos(\theta_2))/2$, which shares the similar behavior as that exhibited in the main text.

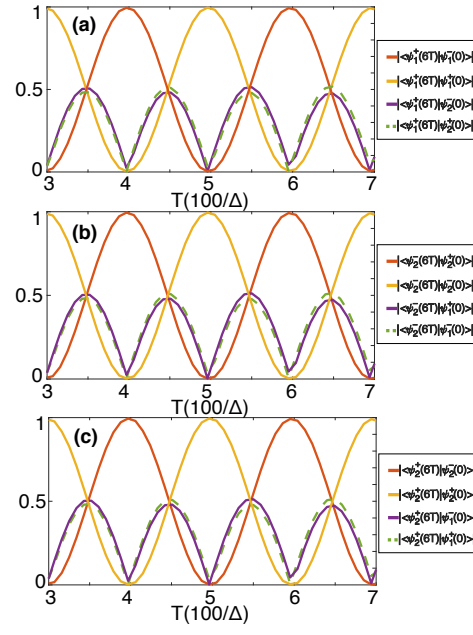


FIG. S6: Numerical simulation of the braiding results as functions of the braiding time-cost T for (a) $\psi_1^+(6T)$, (b) $\psi_2^-(6T)$, and (c) $\psi_2^+(6T)$. They all show similar oscillation behaviors as $\psi_1^-(6T)$ in the main text.